



Overclocking has always been considered a black art where results aren't always as expected. Safety equals information—you'll be well protected after reading this...

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Overclocking as an activity seems to originate from mankind's eternal quest to continuously outdo himself. Automobiles, airplanes and now computers—we seem to like to push performance envelopes.

Quite simply, overclocking is to a PC what turbocharging is to an automobile engine—well not quite but the end result is the same—more performance. Running finely tweaked components at greater speeds than they were designed to run at is however potentially dangerous—to the components (and possibly to your person). Which is why overclocking is an art form for many.

Overclocking involves increasing the working frequencies of all performance impacting components namely the CPU, RAM, GPU and video memory, and other motherboard buses like the Northbridge, Southbridge, and PCI Express bus frequency. A processor that is rated to run at 2.4 GHz and actually running at 2.8 GHz (for example) is said to be “overclocked”.

It's That Simple?

While overclocking its important to remember that the extent of the overclock is largely dependent on the weakest component in your PC. Components that are typically stressed are the motherboard, processor and memory. If you decide to overclock your video card as well then it's the fourth component that is stressed—we say *decide* to because by default *overclocking* as an activity doesn't include VGA overclocking.

You might have an excellent processor, but your choice of RAM may not allow you to overclock too high. Or you may be using a strictly budget motherboard.

Components have to be perfectly matched. In fact, those thinking about overclocking with a value motherboard—think again! While most people buying a PC think of the best processor within their budget and then spend the remainder on a board. Being a potential overclocker you should think differently—buy the best possible motherboard so you can reach higher overlocks with your CPU of choice.

While buying the best board within your budget is important, you don't necessarily have to splurge on the *best processor*. For example—we've got a Core 2 Duo E6600 (default 2.4 GHz) to run at 2.93 GHz which is the default speed of the Core 2 Extreme X6800—a CPU that costs thrice as much. Which means you can reach the same level of performance at one third the price!

What You Need

The first thing you need while building an *overclockers delight* is a good motherboard. Good doesn't necessarily mean costly although as with everything you get what you pay for. Your choice of motherboard will largely determine the extent of overclocking of components possible, and also how easy it is to overclock.

We're looking at an Intel-based PC for building an overclockable system. Intel processors overclock much better, and offer better performance for the buck even at stock speeds.

Processor

The processor is by far the simplest component you'll have to choose. Once again the costliest processor doesn't really necessitate the best overlocks. We're not looking at the highest clock speeds possible (which the best processor would give you). Rather we're looking at the biggest difference between stock speeds and overclocked speeds. Into this ratio also comes the all impor-

tant bang for the buck—overclockers aren't usually uber rich—which is why they overclock in the first place!

As mentioned you don't need the costliest processor. Just to prove our point for this exercise we're taking a Core 2 Duo E6550 (2.33 GHz core, 1333 MHz FSB) as our test subject. This processor costs around Rs 6,500 (street price) and is one of the best buys for an overclocking setup. It can easily reach the speeds of an E6850 when overclocked which is a 3.0 GHz core. For your information this processor has a multiplier of 7 (we'll look at why that is significant later).

If you want to spend a little more we recommend a Core 2 Duo E6750 at rupees 8,400—with a multiplier of 8.

If you're looking at the power of four cores—we recommend a Core 2 Quad Q6600 at Rs 11,400—it overclocks way beyond even a QX6850 (3.00 GHz).

Beyond a point, no amount of tweaking / overvolting will get your clocks higher. This is because of heat, and your CPU temperatures will climb. This is where CPU cooling comes into play. We used Intel's stock cooler for this exercise, but also had a Thermalright Ultra 120 on hand to cool things down in case temperatures got out of hand.

RAM

The choice of memory doesn't really affect the choice of processor but it does affect the choice of motherboard. All Intel chipset motherboards (P35, X38, G33 etc) have a certain fixed number of ratio options between the frequency of the FSB and memory which has to be maintained. These ratios are like 1.5 : 1, 2.0 : 1 and so on and are called *Memory Divider Ratios*. The biggest problem Intel chipset based boards have is the memory divider ratio—simply because overclocking your FSB means automatically increasing the memory clock as well. We call this a problem because most of us are running on value memory—which doesn't overclock well.

In case your PC is running value memory on an Intel chipset based board you may well be limiting your overclocking potential.

NVIDIAs 6xx family of chipsets have a very different approach to overclocking memory and FSB—there's no relation, as they've unlinked each component—no memory dividers plaguing overlocks anymore as the memory frequency and FSB frequency is totally asynchronous.

The choice is simple—if you want to overclock higher with cheaper memory—go for an NVIDIA 680i/650i chipset based motherboard. If you have high quality DDR2 memory and are looking to overclock memory as well then you can go with either Intel or NVIDIA chipset-based boards.

Your choices are limited for *enthusiast grade* memory, unless you can buy it online or have a relative bring it over from the States or Europe. Corsair does have certain high performance kits available locally—their XMS 2 series are basically high frequency, low latency kits—DDR2 800 MHz will cost you nearly triple the price of a regular 2 GB 667 MHz kit.



As far as value goes, we recommend Corsair, Transcend and Kingston (in that order). A 2 GB DDR2 667 MHz kit would cost you between Rs 2,500 (Transcend) and Rs 2,900 (Corsair).

We used Corsair Dominators PC2 8500 rated at 1066 MHz at timings of 5-5-5-15. These sticks use the “overclocker acclaimed” Micron D9 memory chips—the speciality of these chips are in their overclocking potential and memory with these chips is 30 percent costlier than identically clocked memory based on other chips.

Motherboard

If you're looking for an NVIDIA board, we suggest an ASUS P5N32-E SLI; for Intel, we suggest an ASUS P5K Deluxe. Both boards are priced at just below the Rs 15,000. For something cheaper we suggest the Abit IP35-E Off Limits (Intel; Rs 8,500), or the XFX 650i SLI (NVIDIA; Rs 6,500).

The 680i chipset does give a lot more control over various components than the P35-based boards, though we've seen some ASUS boards based on Intel chipsets with detailed BIOS', rich with overclocking options.

For this exercise, we used ASUS' Maximus Formula Limited Edition, based on Intel's latest X38 chipset.

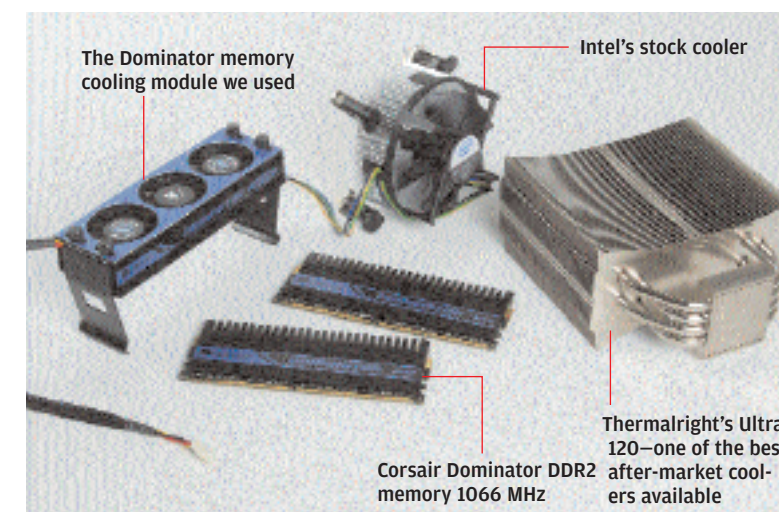
Method To The Madness

Overclocking isn't about madly jacking up frequency figures. Although overclocking your processor will likely cause a bumping up of memory speeds too, let's take this one component at a time.

Processor Overclocking

There are two ways to overclock a processor. The first involves increasing the multiplier which is the figure which when multiplied by the FSB gives us the CPU frequency. For example: 266 MHz FSB x 9 = 2394 MHz (2.4 GHz). This method gives very quick results but doesn't allow one to fine tune, as the frequency jump between even an increment of *one* in the multiplier value would be enormous.

All Intel and AMD processors come with upward locked multipliers so unless you have an Extreme Edition (Intel) or an FX series CPU (AMD)



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